Lesson notes · for you

Sequences and the memory that fades

University, 3

Time
45 min

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- The state is the memory
- Empty steps still multiply it

Activity: In the lab

1. Slider to the left
2. Slider to the right

Discussion

- What would a longer gap do?

Check yourself

- Does the end still know the cue? Write the number.

Try it this week
Write the hidden state after a cue of 1 and one empty step, if memory is one half.

Worksheet

Sequences and the memory that fades

University, 3

Name _____ Class _____ Date _____

1 **Three empty steps follow a cue. Memory is near 0. What is left at the end?**

- ◯ Almost none of the cue
- ◯ The whole cue
- ◯ A new cue the net invented

2 **A recurrent net reads the whole sequence in one look.**

- ◯ True
- ◯ False

3 **Match the word to the meaning.**

Hidden state	• • • •	A small memory times the state, again and again
Cue	• • • •	The symbol that must still matter at the end
Fading	• • • •	What the net carries forward

4 **Order one recurrent step.**

- ☐ Keep the mix as the new state
- ☐ Mix it with the old hidden state
- ☐ Read the next symbol

✎ **What did the empty steps do to the cue?**